Medical Terminology — PYQ Answers

Paper Code: BBA(HM) 201 | BBAHM 201 | MAKAUT, West Bengal

CS/BBACS/EVEN/SEM-2/20000081/2024-2025/187 | Time: 3 Hours | Full Marks: 70 | UPD: 20000081

GROUP A — Very Short Answer Type Questions [1 × 10 = 10]

Answer any ten of the following.

i. What happens when Hypersecretion of thyroid takes place? 1 mark

When **hypersecretion of thyroid hormones (T3 and T4)** occurs, it leads to **Hyperthyroidism** (most commonly *Graves' disease*). Features include: rapid heartbeat (tachycardia), weight loss despite increased appetite, heat intolerance, excessive sweating, anxiety, tremors, diarrhoea, exophthalmos (bulging eyes in Graves' disease), and goitre (enlarged thyroid gland).

ii. LFT stands for? 1 mark

LFT = Liver Function Test. It is a group of blood tests that assess the health and function of the liver. It includes: ALT (Alanine Transaminase), AST (Aspartate Transaminase), ALP (Alkaline Phosphatase), GGT (Gamma-Glutamyl Transferase), Serum Bilirubin (total, direct, indirect), Albumin, and Prothrombin Time (PT).

iii. Name one diagnostic procedure of the gastroenterology department. 1 mark

Colonoscopy — a flexible fibre-optic camera is inserted through the rectum to visualize the entire large intestine (colon). It is used to diagnose colorectal cancer, polyps, inflammatory bowel disease (IBD), and diverticulosis, and allows simultaneous biopsy or polypectomy.

Other acceptable answers: EGD (Upper GI Endoscopy), ERCP, Sigmoidoscopy, Capsule Endoscopy, MRCP.

iv. Define the concept of Medical Terminology. 1 mark

Medical Terminology is a standardized, universally accepted language used by healthcare professionals to accurately describe the human body, its structures, functions, diseases, conditions, and procedures. It is primarily derived from **Greek and Latin** roots and ensures precision and clarity in medical communication, reducing errors and ambiguity across clinical settings.

v. Inflammation of the kidney is known as ____. 1 mark

Inflammation of the kidney is known as **Nephritis** (from Greek: *nephros* = kidney + *itis* = inflammation). The most common form is **Glomerulonephritis** — inflammation of the glomeruli (filtering units), often immune-mediated, presenting with haematuria, proteinuria, oedema, and hypertension.

vi. WNL signifies ____ in prescription. 1 mark

WNL = Within Normal Limits. It is used in prescriptions, clinical notes, and investigation reports to indicate that the patient's findings, test results, or examination outcomes are within the acceptable normal reference range.

vii. Full form of PET Scan. 1 mark

PET Scan = Positron Emission Tomography. It is a nuclear medicine imaging technique that uses a radioactive tracer (commonly FDG — fluorodeoxyglucose) to detect metabolic activity in tissues. Cancer cells consume more glucose and hence appear as "hot spots." Often combined with CT as **PET-CT** for precise anatomical and metabolic localization.

viii. Define cell. 1 mark

A **cell** is the **smallest structural and functional unit of all living organisms**. It is capable of carrying out all the basic processes of life including metabolism, growth, reproduction, and response to stimuli. The human body contains approximately 37 trillion cells. Cells are surrounded by a plasma membrane, contain cytoplasm, and (in eukaryotes) a membrane-bound nucleus housing genetic material (DNA).

ix. The abbreviation "CXR" signifies ____. 1 mark

CXR = Chest X-Ray. It is the most commonly performed radiological investigation in medicine. A standard posteroanterior (PA) CXR is used to evaluate the lungs (pneumonia, TB, COPD, cancer), heart size (cardiomegaly), pleural effusion, pneumothorax, and bony thorax. It uses a low dose of ionizing radiation.

x. What is Appendectomy? 1 mark

Appendectomy (also Appendicectomy) is the **surgical removal of the vermiform appendix**. It is the standard treatment for **acute appendicitis** (inflammation of the appendix). It can be performed as an open surgery (laparotomy) or more commonly via **laparoscopic (keyhole) surgery**. If untreated, appendicitis can lead to perforation and peritonitis.

xi. The word / abbreviation "stat" in prescription means ____. 1 mark

STAT comes from the Latin word *"statim"* meaning **"immediately"** or "at once." In a prescription or clinical order, STAT means the medication or action must be administered or performed without any delay — it is an urgent, emergency directive. Example: "Inj. Adrenaline 0.5 mg IM STAT" in anaphylaxis.

GROUP B — Short Answer Type Questions [5 × 3 = 15]

Answer any three of the following.

1. Write a short note on Thalassaemia. 5 marks

Thalassaemia is an **inherited (autosomal recessive) disorder** of haemoglobin synthesis, characterized by reduced or absent production of one or more globin chains of haemoglobin, leading to chronic haemolytic anaemia.

> **Types:**
>
> - **Alpha-thalassaemia** — defect in alpha-globin chain production; more common in South-East Asia and Africa.
> - **Beta-thalassaemia** — defect in beta-globin chain; most clinically significant form; prevalent in Mediterranean, Middle East, India.

Severity spectrum (Beta-thalassaemia):

- *Thalassaemia minor (trait)* — carrier; mild anaemia; usually asymptomatic
- *Thalassaemia intermedia* — moderate anaemia; may need occasional transfusions

- *Thalassaemia major (Cooley's anaemia)* — severe; requires lifelong regular blood transfusions every 2–3 weeks

Clinical Features: Pallor, fatigue, growth retardation, hepatosplenomegaly (enlarged liver and spleen), skeletal deformities (chipmunk facies — frontal bossing due to marrow expansion), gallstones, iron overload.

Diagnosis: CBC (microcytic hypochromic anaemia), Peripheral blood smear (target cells), Haemoglobin electrophoresis (elevated HbF, reduced/absent HbA), genetic testing.

Treatment: Regular blood transfusions + Iron chelation therapy (deferoxamine, deferasirox) to prevent iron overload. **Curative:** Bone marrow/stem cell transplantation. Prenatal diagnosis via chorionic villus sampling (CVS) or amniocentesis.

2. Write down the names of various body cavities in our body. Mention the organs and parts positioned on those body cavities. 5 marks

The human body has two major cavity systems — the **Dorsal Cavity** and the **Ventral Cavity**.

Major Cavity	Subdivision	Organs Contained
Dorsal Cavity	Cranial cavity	Brain (cerebrum, cerebellum, brainstem), meninges
	Vertebral / Spinal cavity	Spinal cord, spinal nerve roots, meninges, CSF
Ventral Cavity	Thoracic cavity	Heart (in pericardium), lungs (in pleural sacs), trachea, oesophagus, great vessels (aorta, vena cava), thymus
	Abdominal cavity	Stomach, small intestine, large intestine, liver, gallbladder, pancreas, spleen, kidneys, adrenal glands, aorta, inferior vena cava
	Pelvic cavity	Urinary bladder, rectum, sigmoid colon, uterus + ovaries + fallopian tubes (female); prostate gland + seminal vesicles (male)

The thoracic and abdominal cavities are separated by the **diaphragm**. The abdominal and pelvic cavities together are called the **abdominopelvic cavity**.

3. What is Endoscopy? 5 marks

Endoscopy is a minimally invasive diagnostic and therapeutic procedure in which a thin, flexible instrument called an **endoscope** — equipped with a light source and camera — is inserted into the body to visualize internal organs and structures. The term comes from Greek: *endon* (within) + *skopein* (to look).

> **Components of an Endoscope:** Flexible insertion tube | Light source (fibre-optic or LED) | Camera/CCD chip | Working channel (for instruments — biopsy forceps, snare, injector needle)

Types of Endoscopy:

Type	Area Examined	Clinical Use
EGD (Upper GI Endoscopy)	Oesophagus, stomach, duodenum	Peptic ulcers, GERD, gastric cancer
Colonoscopy	Entire colon + terminal ileum	CRC screening, IBD, polyps
Bronchoscopy	Trachea, bronchi	Lung cancer, TB, foreign body
Cystoscopy	Bladder, urethra	Bladder cancer, UTI, stones

Laparoscopy	Abdominal cavity	Endometriosis, appendectomy, cholecystectomy
Arthroscopy	Joints (knee, shoulder)	Ligament tears, meniscus repair

Advantages: No large incision, shorter hospital stay, direct visualization + biopsy in one procedure, real-time images. **Risks:** Bleeding, perforation, infection (rare), adverse reaction to sedation.

4. Write down Medical Terms to define the Endocrine system of the body. 5 marks

The **Endocrine system** comprises glands that secrete hormones directly into the bloodstream. Key medical terms:

Medical Term	Meaning / Relevance to Endocrine System
Endocrinology	Study of the endocrine system and its diseases
Hormone	Chemical messenger secreted by endocrine glands
Hyperglycaemia	High blood glucose — hallmark of Diabetes Mellitus
Hypoglycaemia	Abnormally low blood glucose
Hypothyroidism	Deficient thyroid hormone secretion (low T3/T4)
Hyperthyroidism	Excess thyroid hormone (Graves' disease)
Adrenal cortex / medulla	Cortex secretes cortisol, aldosterone; Medulla secretes adrenaline
Insulinoma	Insulin-secreting tumour of the pancreatic islets
Goitre	Enlargement of the thyroid gland
Acromegaly	Excess Growth Hormone (GH) in adults → enlarged hands, feet, jaw
Gigantism	Excess GH in children → abnormal height
Diabetes insipidus	Deficiency of ADH (antidiuretic hormone); large dilute urine output
Addison's disease	Adrenal insufficiency — low cortisol and aldosterone
Cushing's syndrome	Excess cortisol; moon face, central obesity, buffalo hump, striae
RAAS	Renin-Angiotensin-Aldosterone System — regulates blood pressure and fluid balance

5. Enumerate anatomical divisions with the help of the diagram. 5 marks

The human body is divided anatomically using **body planes** and **regions/quadrants**:

A. Body Planes (used to describe sections):

Plane	Divides Body Into
Sagittal (Midsagittal)	Left and right halves (equal halves = midsagittal)
Frontal (Coronal)	Anterior (front) and posterior (back) portions
Transverse (Axial / Horizontal)	Superior (upper) and inferior (lower) portions
Oblique	Diagonal cut at an angle

B. Abdominopelvic Regions — 9 Regions:

Right Hypochondriac	**Epigastric**	Left Hypochondriac
Right Lumbar	**Umbilical**	Left Lumbar
Right Iliac (Inguinal)	**Hypogastric (Pubic)**	Left Iliac (Inguinal)

C. Abdominopelvic Quadrants — 4 Quadrants:

RUQ Right Upper Quadrant (liver, gallbladder, right kidney)	**LUQ** Left Upper Quadrant (stomach, spleen, left kidney)
RLQ Right Lower Quadrant (appendix, right ovary, cecum)	**LLQ** Left Lower Quadrant (sigmoid colon, left ovary, descending colon)

GROUP C — Long Answer Type Questions [15 × 3 = 45]

Answer any three of the following.

7(a). Describe the digestive system and its mechanism. 9 marks

Introduction: The digestive system is a complex series of organs and glands that processes food, absorbs nutrients essential for energy, growth and cell repair, and eliminates waste products from the body. It extends from the mouth to the anus — approximately 9 metres in total length.

Organs of the Digestive System:

- **Alimentary canal (GI tract):** Oral cavity → Pharynx → Oesophagus → Stomach → Small intestine (Duodenum, Jejunum, Ileum) → Large intestine (Cecum, Colon, Rectum, Anal canal)
- **Accessory organs:** Salivary glands, Liver, Gallbladder, Pancreas

Mechanism — Stages of Digestion:

Ingestion → Mechanical Digestion → Chemical Digestion → Absorption → Defaecation

Organ	Function	Key Enzymes/Secretions
Mouth	Mechanical (chewing) + initial chemical digestion of starch	Salivary amylase (breaks starch → maltose)
Oesophagus	Peristalsis transports bolus to stomach	Mucus (lubrication only)
Stomach	Churns food into chyme; protein digestion begins	HCl (pH 1.5–3.5), Pepsin (protein → peptides), Gastric lipase
Small intestine (Duodenum)	Main site of chemical digestion; bile + pancreatic juice mix with chyme	Pancreatic amylase, Lipase, Trypsin/Chymotrypsin; Bile (fat emulsification)

Small intestine (Jejunum/Ileum)	Primary site of nutrient **absorption** (via villi and microvilli)	Brush border enzymes: maltase, lactase, sucrase
Large intestine	Water and electrolyte absorption; formation and storage of faeces	Gut flora (vitamin K, B12 synthesis)
Liver	Produces bile; metabolises nutrients absorbed from portal blood	Bile acids, bilirubin
Pancreas	Produces digestive enzymes (exocrine) + insulin/glucagon (endocrine)	Pancreatic juice (amylase, lipase, proteases, bicarbonate)
Gallbladder	Stores and concentrates bile from liver; releases it into duodenum	Bile (contains bile salts, cholesterol, bilirubin)

Conclusion: The digestive system works in a coordinated manner — mechanical grinding breaks food into smaller pieces, enzymatic action converts macronutrients (carbohydrates, proteins, fats) into absorbable units (glucose, amino acids, fatty acids), which are then absorbed primarily in the small intestine and transported to the liver via the portal vein for further processing. Waste is eliminated through the large intestine.

7(b). What do you understand by Peptic Ulcer? 3 marks

A **Peptic Ulcer** is an open sore (erosion) in the lining of the stomach (**gastric ulcer**) or the upper part of the small intestine (**duodenal ulcer**), caused when the protective mucus layer is damaged and stomach acid erodes the underlying tissue.

Causes: H. pylori (Helicobacter pylori) bacterial infection (most common — ~80%), excessive NSAID (aspirin, ibuprofen) use, smoking, alcohol, stress (physiological in critically ill patients — Curling's, Cushing's ulcers).

Symptoms: Burning/gnawing epigastric pain (typically relieved by food in duodenal ulcer; worsened by food in gastric ulcer), nausea, bloating, haematemesis (vomiting blood), melaena (black tarry stools — indicates upper GI bleed).

Diagnosis: Upper GI endoscopy (EGD) — gold standard; H. pylori testing (urea breath test, rapid urease test, stool antigen test).

Treatment: H. pylori eradication — Triple therapy (PPI + Clarithromycin + Amoxicillin × 14 days). Stop NSAIDs. Proton Pump Inhibitors (PPIs) — omeprazole, pantoprazole. Antacids, H2 blockers.

7(c). What is Dyspepsia? 3 marks

Dyspepsia (commonly known as indigestion) is a **symptom complex** characterized by persistent or recurring upper abdominal discomfort or pain. It is not a single disease but a constellation of symptoms arising from the upper GI tract.

Symptoms: Epigastric pain or burning, early satiety (feeling full quickly), post-prandial fullness, belching, bloating, nausea, and occasionally vomiting.

Types:

- *Functional (Non-ulcer) Dyspepsia* — no structural cause found on investigation; most common (60–70%); related to altered gut motility, visceral hypersensitivity
- *Organic Dyspepsia* — due to an identifiable cause: peptic ulcer, GERD, gastric cancer, H. pylori, medications (NSAIDs, antibiotics)

Investigation: Upper GI endoscopy (especially if alarm features: weight loss, dysphagia, haematemesis, >55 years), H. pylori testing. **Treatment:** Lifestyle modification (small frequent meals, avoid spicy/fatty food, alcohol, smoking), antacids, PPIs, H2 blockers, prokinetics (domperidone), H. pylori eradication if positive.

8(a). Name any two abbreviations used by doctors in Prescription. 3 marks

Two commonly used abbreviations in prescriptions:

- **BD (Bis in die)** — Latin for "twice a day." The medication must be taken two times per day (e.g., Tab. Metformin 500 mg BD PC — take twice daily after meals).
- **STAT (Statim)** — Latin for "immediately." The dose must be administered at once without delay. Used in emergency situations (e.g., Inj. Hydrocortisone 200 mg IV STAT in anaphylaxis).

Other acceptable answers: OD, TDS, HS, SOS, PRN, PO, IM, IV, AC, PC, NPO, WNL, etc.

8(b). Write a short note on Nuclear Lungs Scanning and Pulmonary artery angiography. 10 marks

A. Nuclear Lung Scanning (Ventilation-Perfusion Scan / V/Q Scan)

Introduction: Nuclear lung scanning, specifically the **Ventilation-Perfusion (V/Q) Scan**, is a nuclear medicine imaging technique used to evaluate airflow (ventilation) and blood flow (perfusion) in the lungs.

Radiotracer inhaled (V scan) + Radiotracer injected IV (Q scan) → Gamma camera images → V/Q mismatch analysis

- **Ventilation (V) phase:** Patient inhales radioactive gas (Xenon-133 or Technetium-99m DTPA aerosol); gamma camera captures lung aeration
- **Perfusion (Q) phase:** Technetium-99m labelled macroaggregates of albumin (MAA) injected intravenously; particles lodge in pulmonary capillaries proportional to blood flow

Clinical Use: Primary indication = **Pulmonary Embolism (PE)** diagnosis. In PE, a segment of lung is ventilated (air reaches it) but NOT perfused (blocked artery prevents blood flow) — this is a *V/Q mismatch* pattern, highly suggestive of PE.

Advantages over CTPA: Lower radiation dose, no iodinated contrast (suitable for patients with renal failure or contrast allergy), safe in pregnancy (lower foetal radiation than CTPA). **Limitations:** Less specific than CTPA; requires nuclear medicine facility.

B. Pulmonary Artery Angiography

Introduction: Pulmonary artery angiography (pulmonary angiography) is an **invasive diagnostic procedure** that provides direct X-ray visualization of the pulmonary arterial tree after injection of iodinated contrast dye through a catheter advanced into the pulmonary artery.

Femoral/jugular vein access → Catheter → right heart → pulmonary artery → Contrast injection → Fluoroscopic/digital X-ray images

Clinical Indications:

- Diagnosis of **Pulmonary Embolism (PE)** — historically the gold standard (now largely replaced by CTPA)

- Evaluation of **Chronic Thromboembolic Pulmonary Hypertension (CTEPH)** before pulmonary endarterectomy (PEA)
- Assessment of pulmonary arteriovenous malformations (AVM)
- Evaluation of pulmonary hypertension severity

Procedure: Performed in a cardiac catheterization lab under local anaesthesia + sedation. A Swan-Ganz catheter is advanced from peripheral vein → right atrium → right ventricle → pulmonary artery. Pressure measurements are also obtained (right heart catheterization combined). Contrast injected and serial X-ray images captured.

Findings in PE: Intraluminal filling defect (clot appears as dark area within contrast-filled artery), cut-off sign (abrupt termination of vessel), oligaemia (zone of decreased vascularity distal to clot).

Risks: Arrhythmias, contrast allergy, bleeding at catheter site, right heart perforation (rare).
Contraindications: Severe contrast allergy, uncorrected coagulopathy, severe pulmonary hypertension.

Conclusion: While CTPA has largely replaced diagnostic pulmonary angiography for PE, it remains essential for CTEPH assessment and interventional procedures. The V/Q scan remains a valuable non-invasive alternative, especially in patients where radiation or contrast must be minimized.

9. Mention the purpose of Nuclear Lung Scanning and Pulmonary Artery Angiography. 4 marks

Purpose of Nuclear Lung Scanning (V/Q Scan):

- Primary purpose: **Diagnosis of Pulmonary Embolism (PE)** — detects V/Q mismatch
- Pre-operative lung function assessment (pneumonectomy planning)
- Evaluation of lung transplant perfusion post-operatively
- Alternative to CTPA when contrast dye is contraindicated or in pregnancy

Purpose of Pulmonary Artery Angiography:

- Gold standard for PE diagnosis (now largely replaced by CTPA)
- Essential assessment before **pulmonary endarterectomy (PEA)** in CTEPH — maps the exact extent and location of chronic clot burden
- Diagnosis and treatment of pulmonary arteriovenous malformations (coil embolization via same catheter)
- Hemodynamic assessment in pulmonary arterial hypertension (PAH) — measures pulmonary artery pressure, pulmonary vascular resistance, cardiac output

10. Draw and level the structure of a cell with every cell organelle. 15 marks

The Cell — Structure with All Organelles:

> Note: Draw a large oval (cell outline) with the following labelled structures inside. Below is the descriptive diagram with all organelles and their functions for the written answer.

CELL DIAGRAM — Descriptive Representation

Outer boundary: Cell Membrane (Plasma Membrane) — phospholipid bilayer
Internal fluid medium: Cytoplasm — cytosol + all organelles

Nucleus (central control):
├─ Nuclear envelope (double membrane)

├─ Nucleolus (ribosome synthesis)
└─ Chromatin / DNA

Organelles in Cytoplasm:
├─ Mitochondria (ATP production — powerhouse)
├─ Rough ER (studded with ribosomes — protein synthesis)
├─ Smooth ER (no ribosomes — lipid synthesis, detox)
├─ Ribosomes (free or ER-bound — site of protein synthesis)
├─ Golgi Apparatus / Golgi body (packages + ships proteins — post office of cell)
├─ Lysosomes (digestive organelles — "suicide bags"; contain hydrolytic enzymes)
├─ Vacuoles (storage organelles — food, water, waste)
├─ Centrioles (paired structures near nucleus — organize spindle during cell division; absent in mature neurons)
├─ Cytoskeleton (microfilaments, intermediate filaments, microtubules — shape + movement)
└─ Peroxisomes (break down fatty acids + hydrogen peroxide using catalase)

Organelle	Function	Analogy
Cell membrane	Controls what enters/exits the cell (selective permeability)	Security gate
Nucleus	Contains DNA; directs all cell activities	Control room / Brain
Nucleolus	Produces rRNA; site of ribosome assembly	Factory within factory
Mitochondria	Produces ATP via aerobic cellular respiration	Powerhouse / Power plant
Ribosome	Translates mRNA into proteins	Protein factory
Rough ER	Synthesis and transport of proteins (secretory)	Assembly line
Smooth ER	Lipid/steroid synthesis; drug detoxification (liver cells)	Chemical lab
Golgi Apparatus	Modifies, sorts, packages proteins for secretion or internal use	Post office / Courier
Lysosome	Intracellular digestion; destroys old organelles (autophagy)	Recycling plant / Suicide bag
Vacuole	Stores food, water, waste, ions	Storage tank
Centriole	Forms aster fibres and spindle during mitosis/meiosis	Construction crew
Cytoskeleton	Maintains cell shape; assists in cell movement and division	Skeleton/scaffold
Peroxisome	Oxidises fatty acids; breaks down H_2O_2 using catalase	Detox centre

11. Write and describe ten Medical Terms used to refer to different disorders of the Nervous System. 10 marks

#	Medical Term	Definition
1	**Cerebrovascular Accident (CVA) / Stroke**	Sudden neurological deficit caused by interruption of blood supply to the brain; ischaemic (clot) or haemorrhagic (bleed); emergency requiring immediate treatment ("time is brain")
2	**Epilepsy / Seizure disorder**	Chronic neurological condition characterized by recurrent unprovoked seizures (abnormal, excessive neuronal firing); classified as focal or generalized; treated with antiepileptic drugs (AEDs)
3	**Meningitis**	Inflammation of the meninges (protective membranes of brain and spinal cord); causes: bacterial (Neisseria meningitidis), viral, fungal; presents with headache,

		neck stiffness (meningism), photophobia, fever
4	**Encephalitis**	Inflammation of the brain parenchyma itself; usually viral (Herpes simplex, rabies, Japanese encephalitis); symptoms: altered consciousness, fever, seizures, focal neurological deficits
5	**Multiple Sclerosis (MS)**	Autoimmune demyelinating disease of CNS; immune system attacks myelin sheaths; symptoms: vision loss, weakness, spasticity, ataxia, cognitive changes; relapsing-remitting pattern
6	**Parkinson's Disease**	Progressive neurodegenerative disorder; dopamine depletion in substantia nigra; classic triad: resting tremor, rigidity, bradykinesia; also postural instability; treated with levodopa/carbidopa
7	**Alzheimer's Disease**	Most common cause of dementia; progressive loss of memory, language, and cognition; amyloid plaques + neurofibrillary tau tangles; no cure; managed with acetylcholinesterase inhibitors (donepezil)
8	**Neuropathy (Peripheral Neuropathy)**	Damage or disease affecting peripheral nerves; causes: diabetes (diabetic neuropathy most common), alcohol, B12 deficiency, Guillain-Barré syndrome; symptoms: tingling, burning, numbness, weakness in hands/feet
9	**Hydrocephalus**	Abnormal accumulation of cerebrospinal fluid (CSF) in the brain ventricles, increasing intracranial pressure (ICP); types: communicating and non-communicating; treated with VP (ventriculoperitoneal) shunt
10	**Trigeminal Neuralgia (Tic Douloureux)**	Severe, sudden, electric shock-like facial pain along the distribution of the trigeminal nerve (CN V); often triggered by eating, speaking, or touching the face; treated with carbamazepine or microvascular decompression surgery

12. Break down the words Myocarditis, Pericarditis, Endocarditis, Cardiologist and Cardiomegaly with respect to Prefix, Root and Suffix. 5 marks

Medical Term	Prefix	Root Word	Suffix	Meaning
Myocarditis	Myo- (muscle)	Card/o (heart)	-itis (inflammation)	Inflammation of the heart muscle (myocardium); caused by viral infection (Coxsackievirus), autoimmune disease; can lead to heart failure
Pericarditis	Peri- (around)	Card/o (heart)	-itis (inflammation)	Inflammation of the pericardium (outer sac surrounding the heart); causes: viral, autoimmune (SLE), post-MI (Dressler's syndrome); presents with sharp chest pain relieved by leaning forward
Endocarditis	Endo- (within/inside)	Card/o (heart)	-itis (inflammation)	Inflammation of the inner lining of the heart (endocardium), particularly affecting the heart valves; infective endocarditis (IE) caused by S. aureus, S. viridans; treated with 4–6 weeks of IV antibiotics
Cardiologist	Cardi/o (heart)	Log/o (study of)	-ist (specialist)	A medical specialist who studies, diagnoses, and treats diseases and conditions of the heart and blood vessels; subspecialties:

				interventional, electrophysiology, heart failure
Cardiomegaly	Cardi/o (heart)	Megal/o (large)	-y (condition of)	Abnormal enlargement of the heart, often due to heart failure, hypertension, cardiomyopathy, or valvular disease; detected on CXR (cardiothoracic ratio > 0.5) or echocardiography

Pattern to remember: All 5 terms contain *cardi/o* (from Greek kardia = heart). The prefixes (Myo, Peri, Endo) locate the anatomical layer — from outside inward: Pericardium → Myocardium → Endocardium.